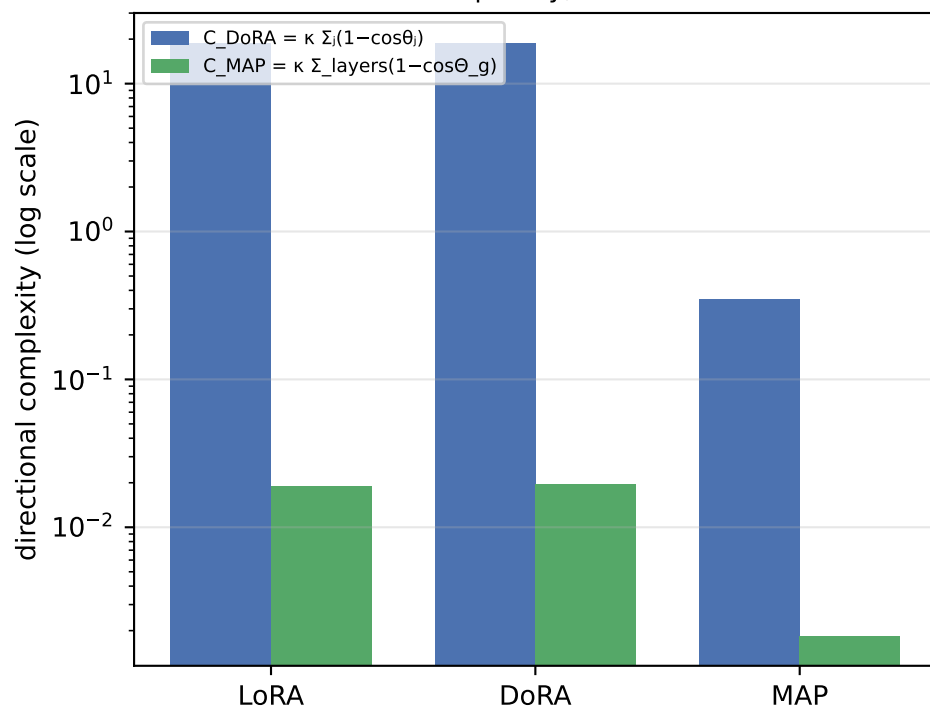


Directional Updates: LoRA vs DoRA vs MAP on RoBERTa-base / RTE

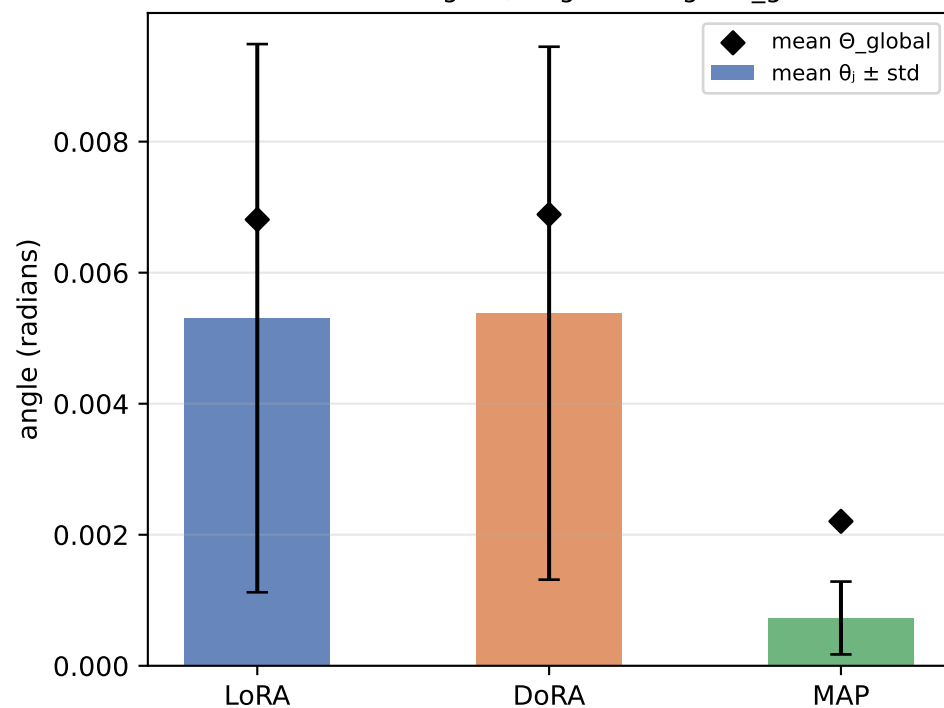
Fine-tuning metrics and aggregate directional complexity

Method	Eval acc	Trainable params	C DoRA (Σ local)	C MAP (global)	mean θ_j (rad)	mean Θ_g (rad)	mean s/d
LoRA	66.06%	1,919,234	18.93	0.019	0.0053	0.0068	0.123
DoRA	67.15%	2,002,178	18.87	0.019	0.0054	0.0069	0.123
MAP	54.15%	1,919,378	0.35	0.002	0.0007	0.0022	0.000

Local (sum) vs global (single rotation)
complexity, $\kappa=10$

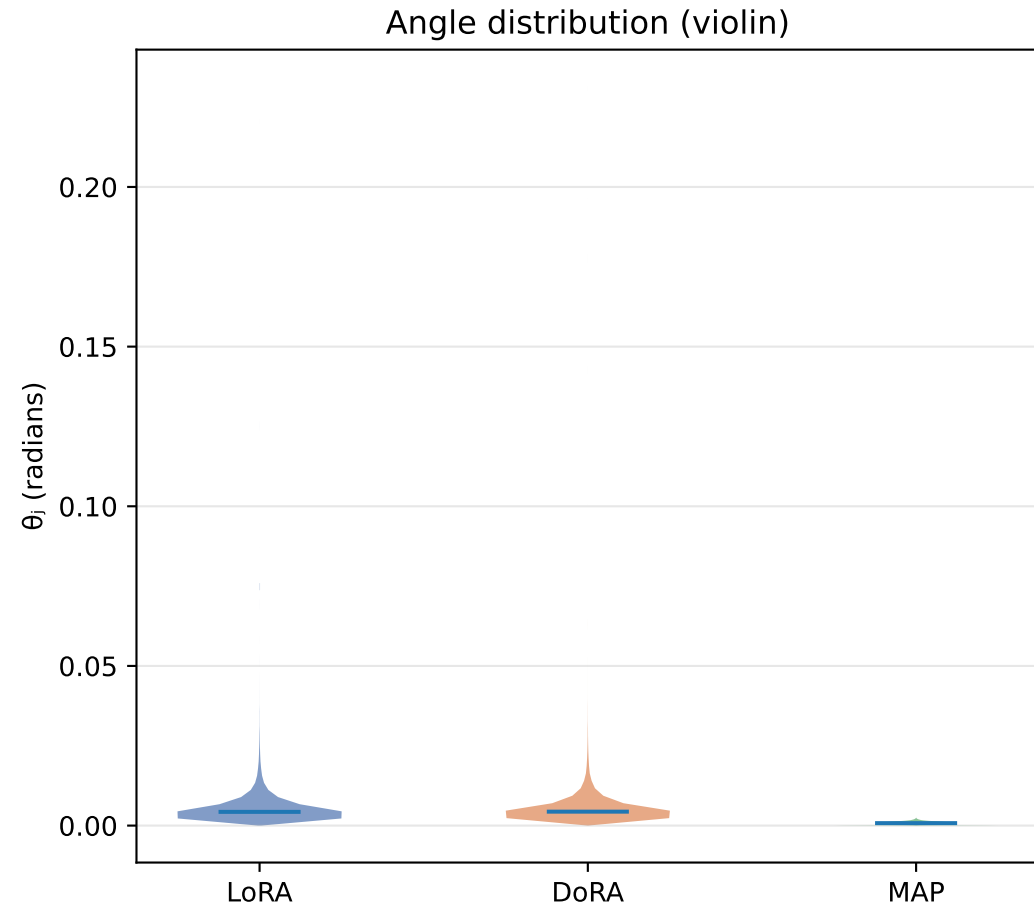
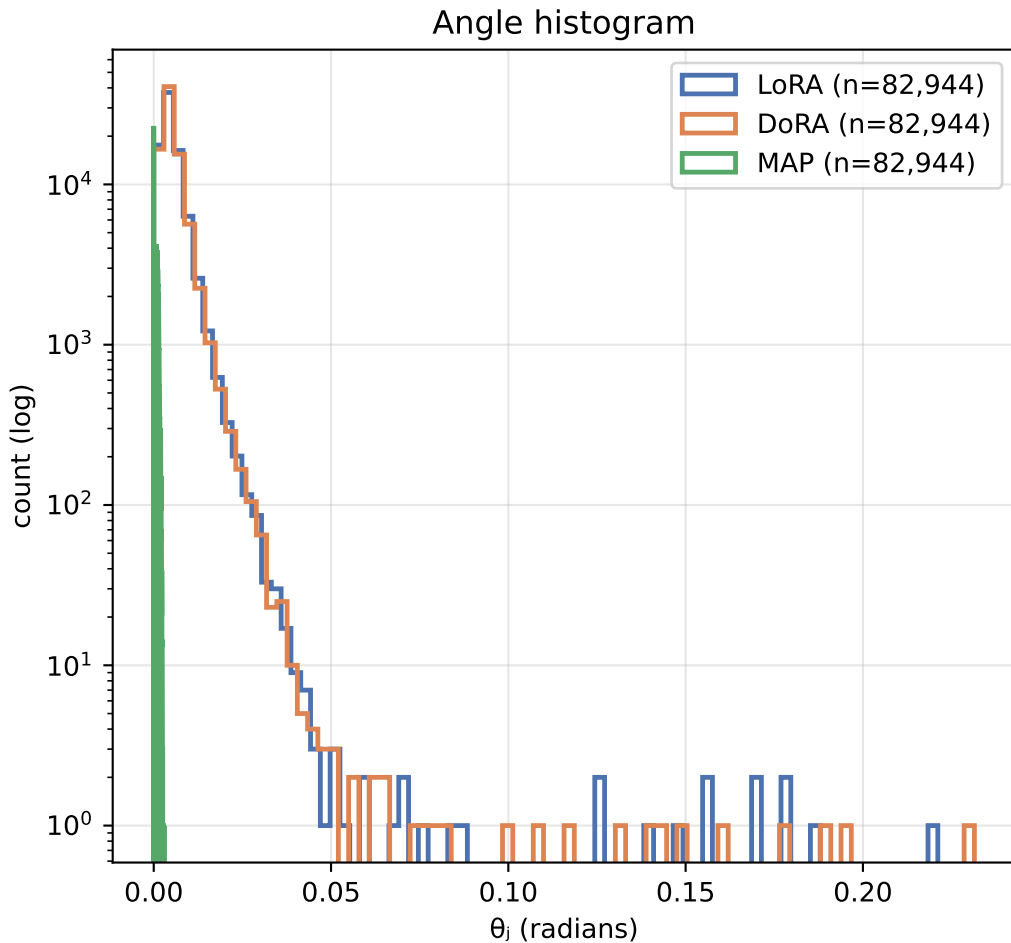


Per-column angle θ_j vs global angle Θ_{global}



Θ_{global} is the single rotation of the whole flattened matrix (MAP view); θ_j are the per-column rotations (DoRA view).
Exact identity (Thm 3.10): $1 - \cos \Theta_{\text{global}} = \text{mean}_j (1 - \cos \theta_j) \rightarrow \text{DoRA sums what MAP averages.}$

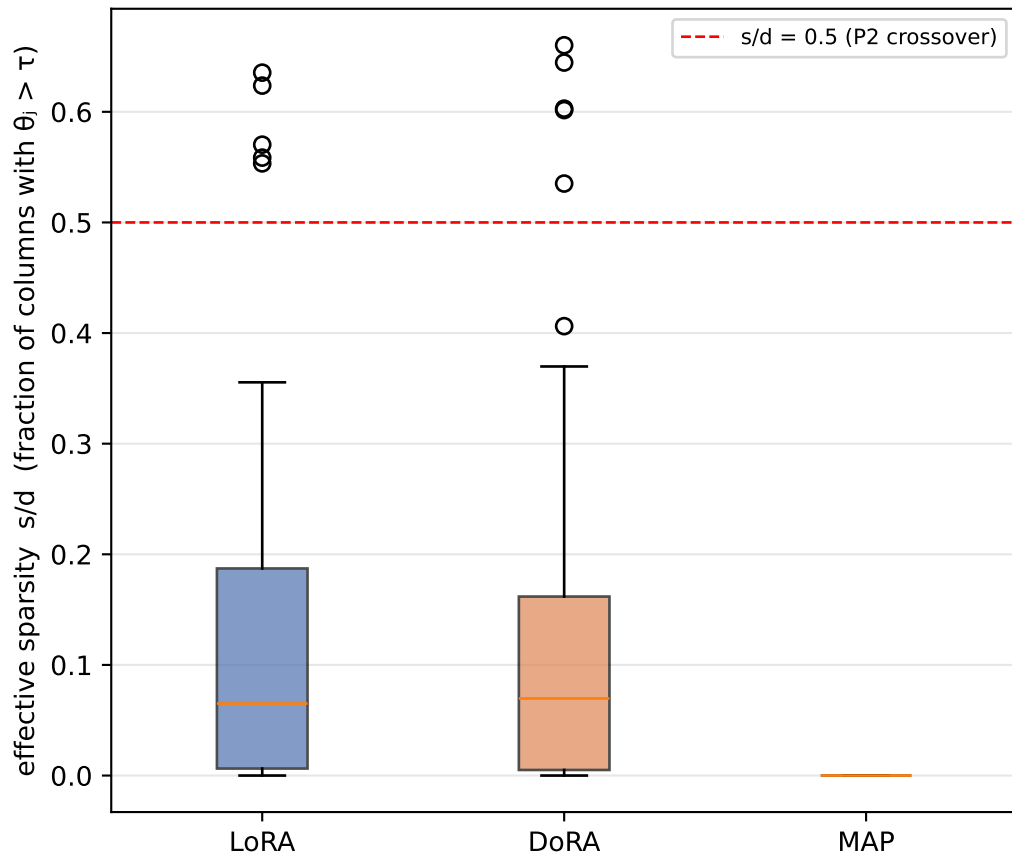
Distribution of per-column directional change θ_j (pooled over all target layers)



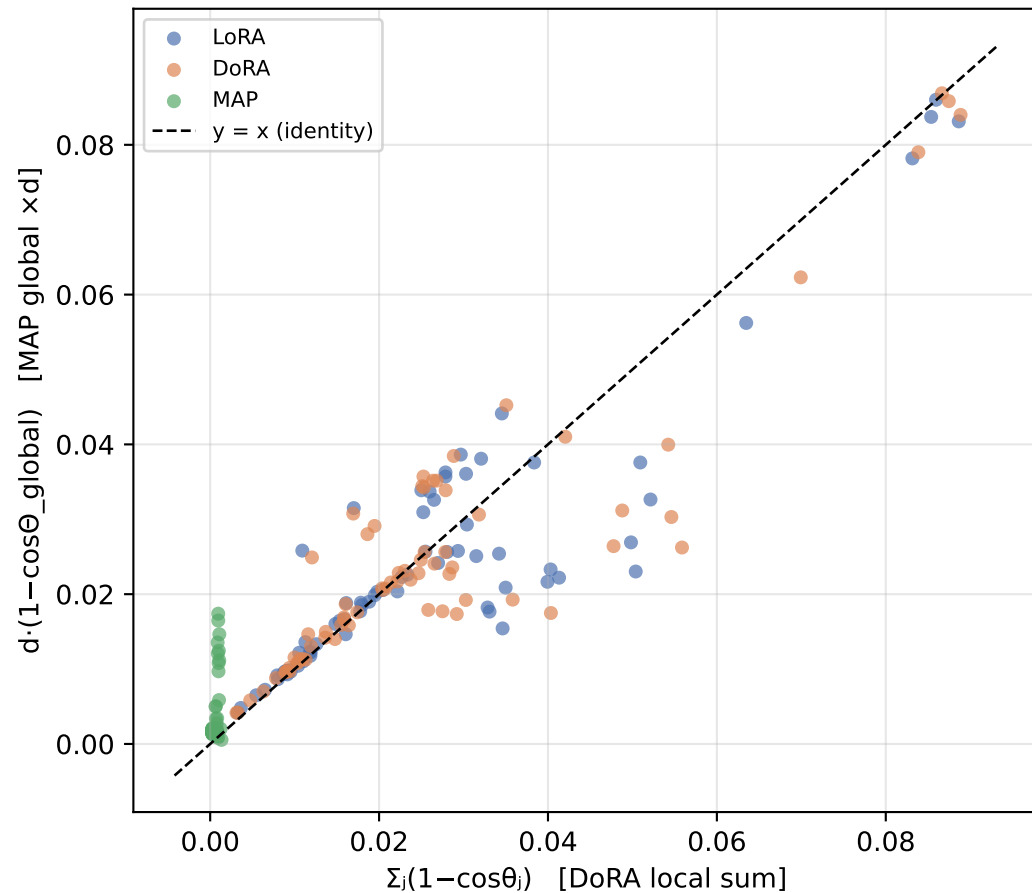
A long right tail = a few columns rotate a lot (sparse/localized update). A concentrated body near 0 = most columns barely move.

Sparsity of directional updates and the DoRA $\leftrightarrow$ MAP identity

Per-layer sparsity ($\tau=0.01$ rad)



DoRA local sum vs MAP global $\times d$ (Thm 3.10 relation, approx: Θ_{global} uses raw — not unit — column flattening)



Where the rotation happens: attention vs MLP, and across depth

